Lecture 18:
Data Visualization (Principles)

INFO 1100: Introduction to Programming

there are many ways to analyze data

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one we’ve overlooked thus far is
visualization

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Python allows us to create a myriad of specialized charts!

there are many ways to analyze data

one we've overlooked thus far is ***visualization***

Python allows us to create a myriad of specialized charts!

but first, we should learn how to make ***good*** visualizations

What makes a ***good*** visualization?

What makes a ***good*** visualization?

We’ll try to answer this!

but first…

but first…

why visualize data?

Why visualize data?

Why visualize data?

Let’s consider an example

Why visualize data?

Let's consider an example

$\mu_x = 9$ $\quad \mu_y = 7.5$ $\quad s_x^2 = 11$ $\quad s_y^2 = 4.13$ $\quad r = 0.81$

Why visualize data?

Let's consider an example

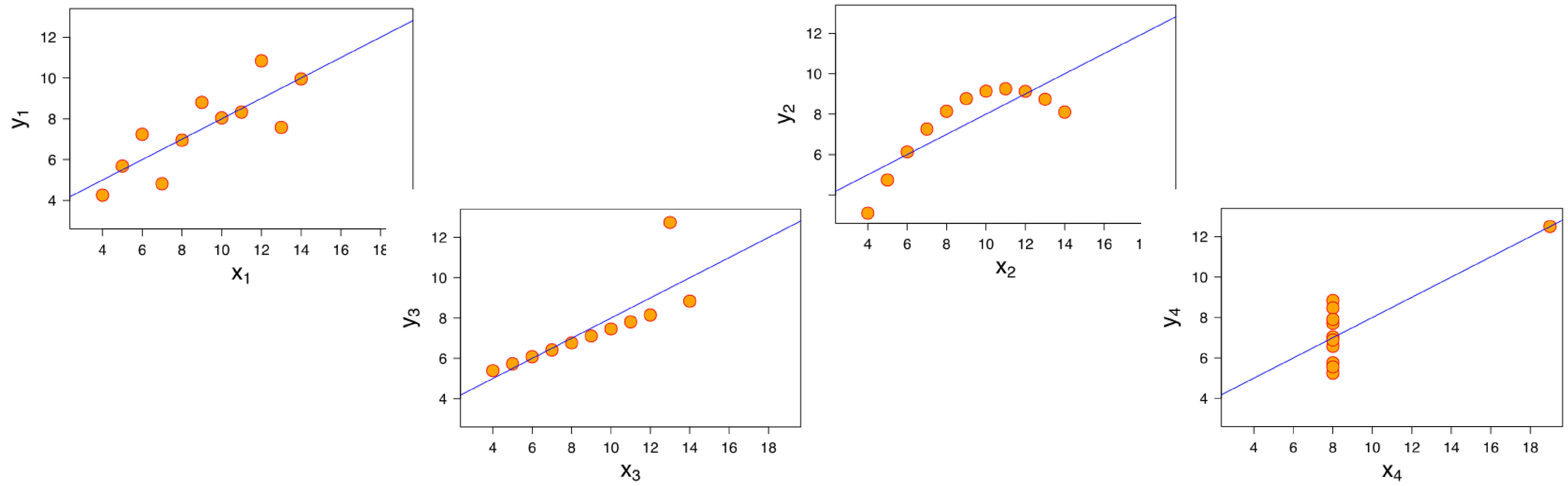
$\mu_x = 9$ $\mu_y = 7.5$ $s_x^2 = 11$ $s_y^2 = 4.13$ $r = 0.81$

What does this data look like?

Why visualize data?

Let's consider an example

$\mu_x = 9$ $\mu_y = 7.5$ $s_x^2 = 11$ $s_y^2 = 4.13$ $r = 0.81$



Why visualize data?

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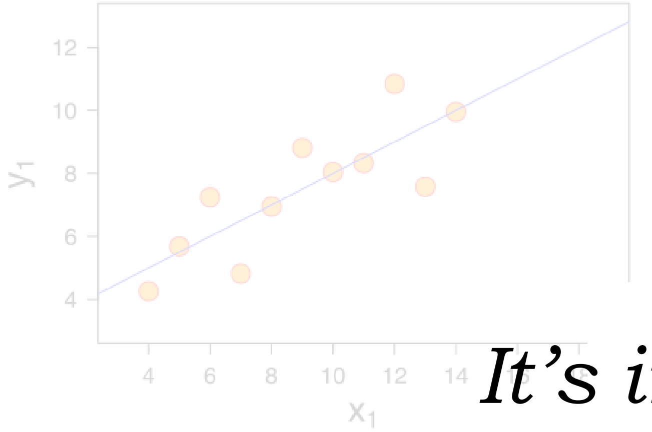
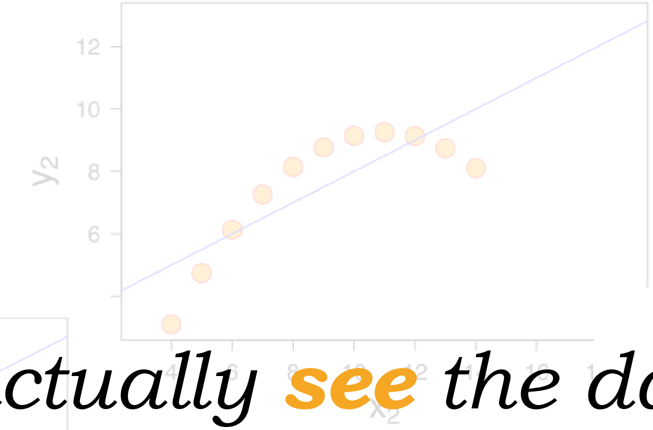
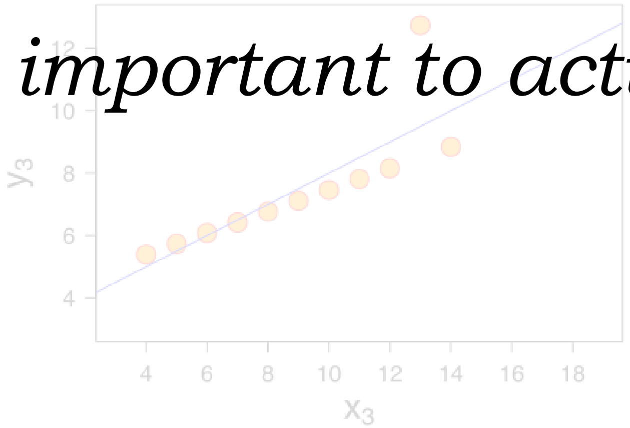
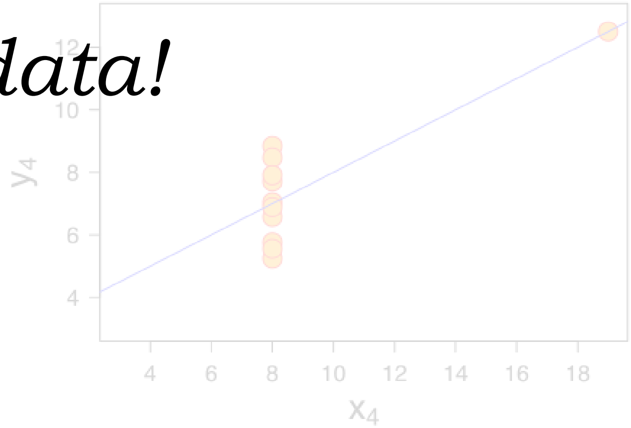




*It's important to actually **see** the data!*





now, to figure out what makes ***good*** visualizations…

now, to figure out what makes ***good*** visualizations…

what makes figures ***bad***?

What makes figures bad?

What makes figures bad?

aesthetic

What makes figures bad?

aesthetic

it looks bad!

What makes figures bad?

aesthetic

What makes figures bad?

aesthetic

there's so much going on!

What makes figures bad?

aesthetic

there's so much going on!

and it doesn't look very nice...

What makes figures bad?

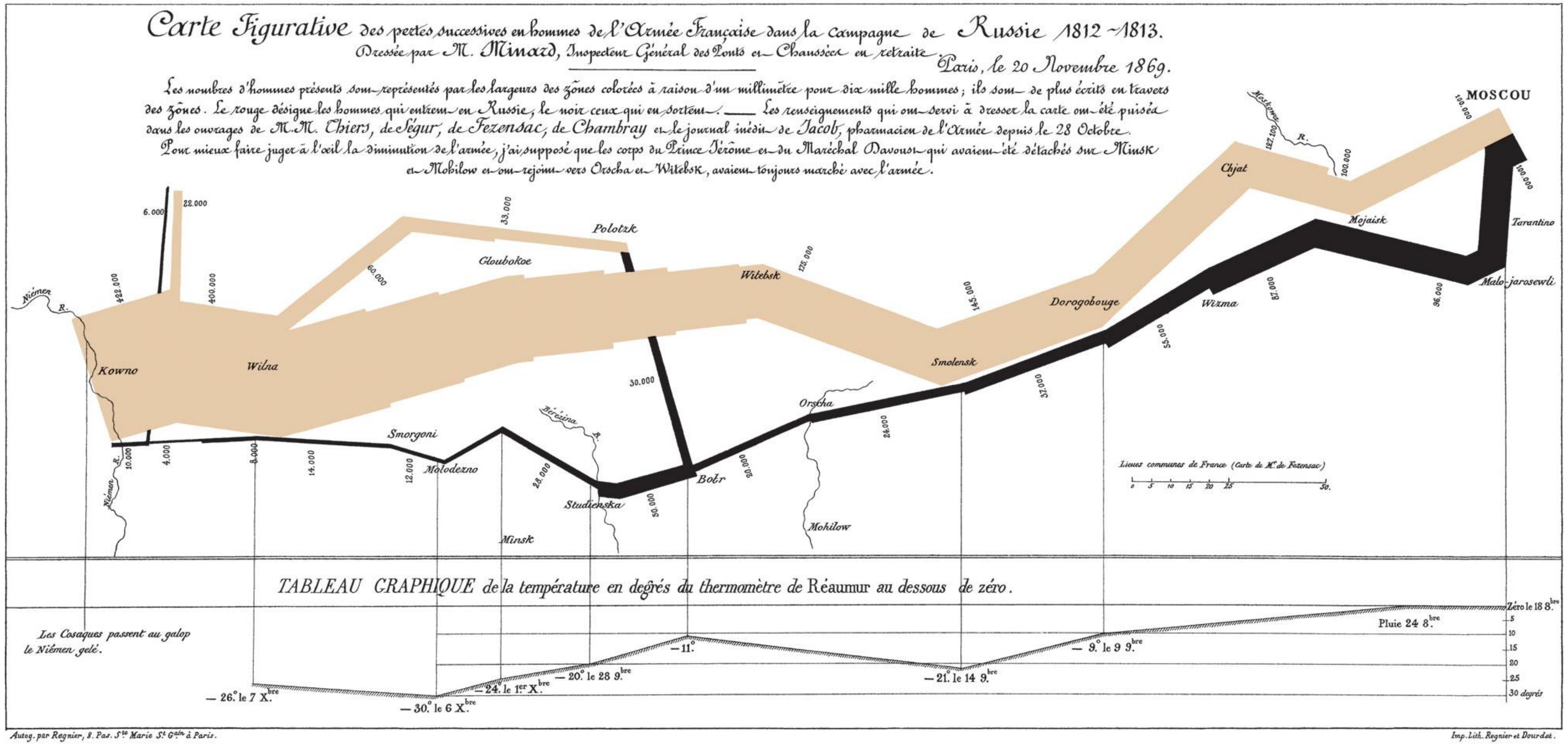
aesthetic

"… the well-designed presentation of interesting data [is] a matter of substance, of statistics, and of design…"

Edward Tufte

What makes figures bad?

aesthetic



What makes figures bad?

aesthetic

how do we make figures this good?

What makes figures bad?

aesthetic

how do we make figures this good?

we usually can't…

What makes figures bad?

aesthetic

often the recommendation is:

What makes figures bad?

aesthetic

often the recommendation is:

maximize the data-to-ink ratio

What makes figures bad?

aesthetic

often the recommendation is:

maximize the data-to-ink ratio

What makes figures bad?

aesthetic

often the recommendation is:

maximize the data-to-ink ratio*

*** you can overdo it!**

What makes figures bad?

aesthetic

seek a balance between simple
and engaging

What makes figures bad?

substantive

What makes figures bad?

substantive

the data is bad!

What makes figures bad?

substantive

your figure is only as
good as your data

What makes figures bad?

substantive

your figure is only as
good as your data

always carefully check your data

What makes figures bad?

perceptual

What makes figures bad?

perceptual

it’s hard to read!

What makes figures bad?

perceptual

we refer to the ways data is
visualized as encodings or channels

What makes figures bad?

perceptual

we refer to the ways data is
visualized as encodings or channels

some are more or less easy for us to interpret

Perception

Perception

a lot of research studies what we're good at interpreting

Perception

essentially: our perceptual cortex is optimized for some tasks

Perception

say we want to estimate or compare
values of data points in a figure

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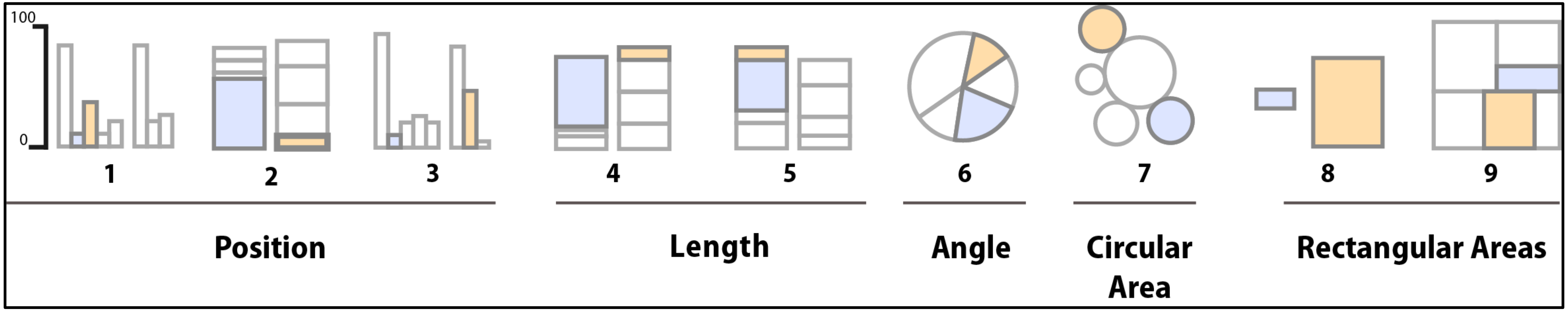
values of data points in a figure

we might choose one of these chart types



Perception

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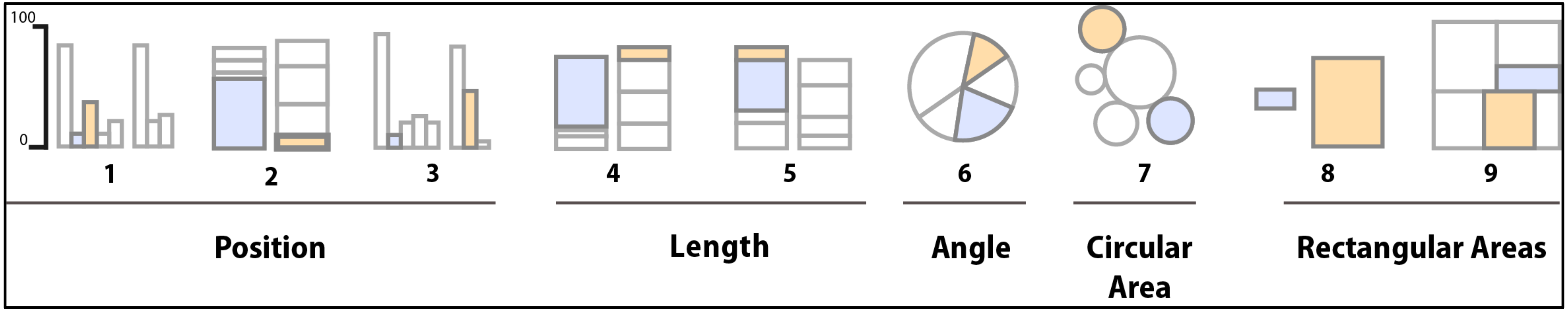

research* tells us charts further to the left are easier to interpret

Heer and Bostock (2010)

Perception

more broadly, for comparing ordered data…

Perception

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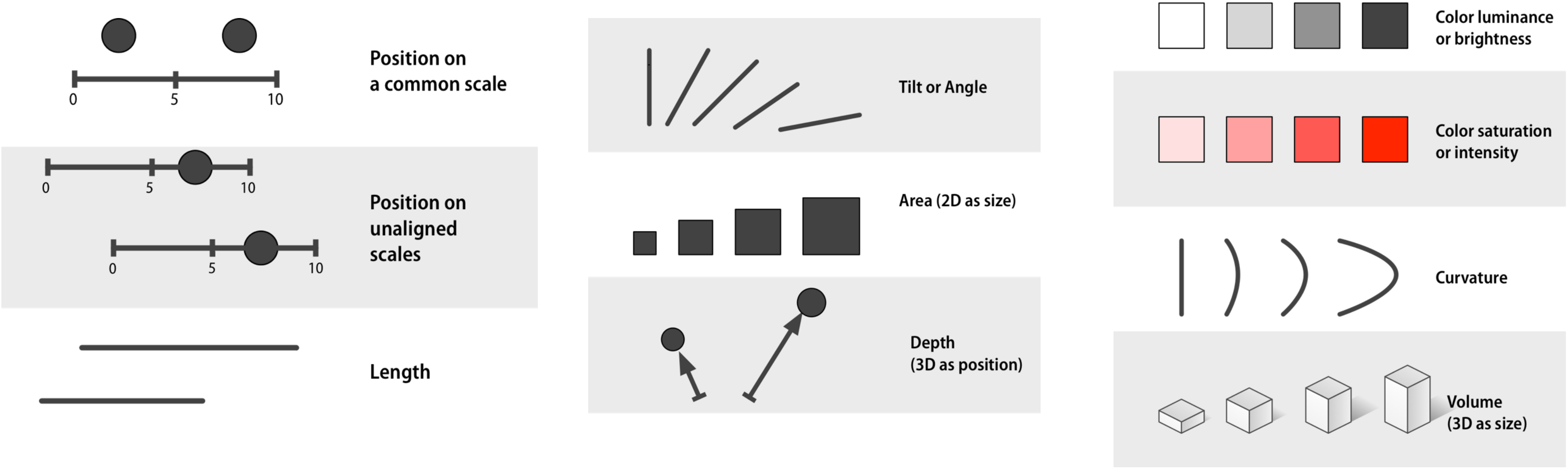
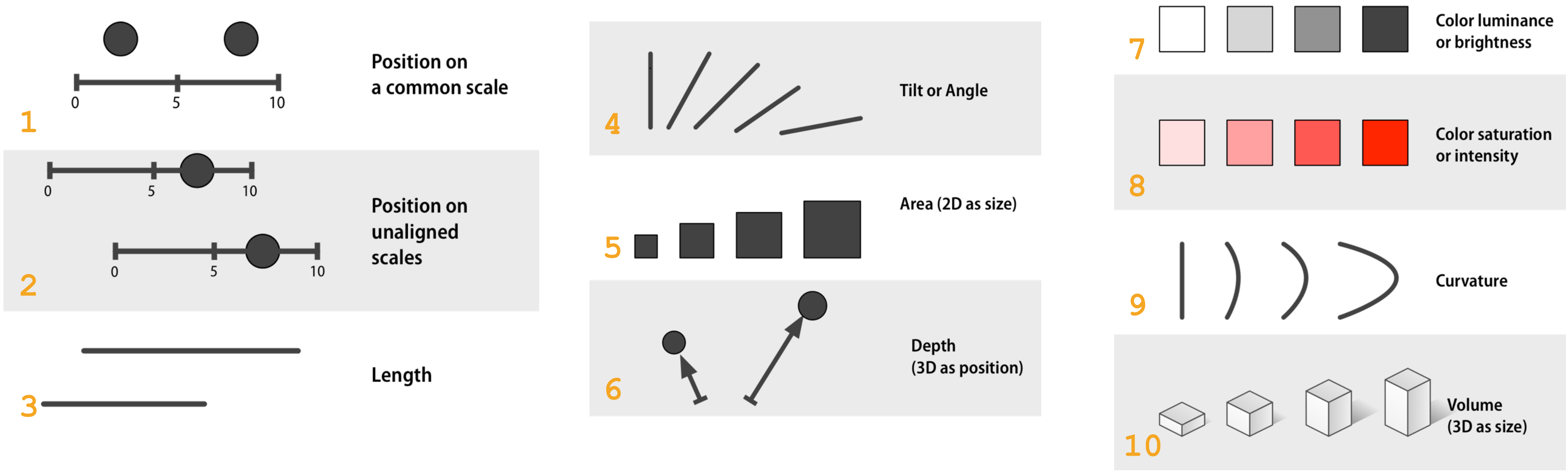


Perception

more broadly, for comparing ordered data...



research can rank how effective these channels are

Perception

and for comparing unordered data…

Perception

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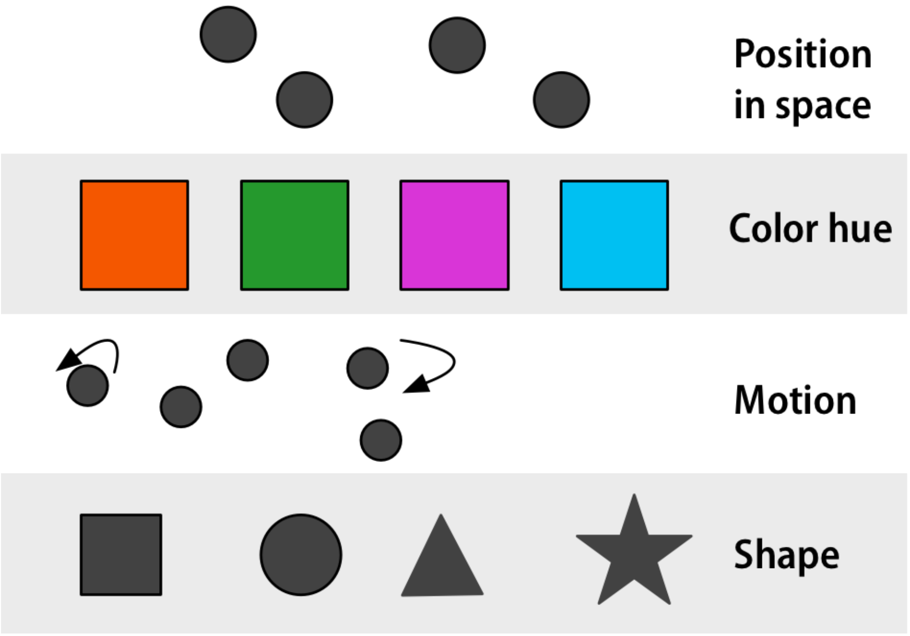
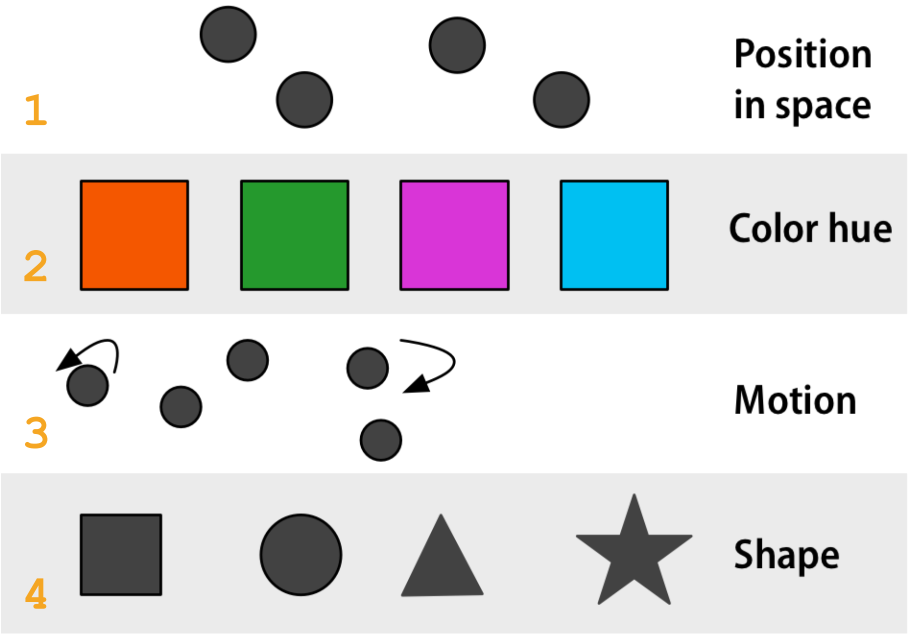


Perception

and for comparing unordered data…



research gives us these rankings

Perception

so…

Perception

so…
what does this mean for our charts?

Perception

use the most accurate channels for the most important data!

Perception

avoid visualizations that rely on perceptually challenging tasks

Perception

avoid visualizations that rely on perceptually challenging tasks

comparing angles (4) and 2D areas (5) are harder than position on a common scale (1)

Perception

avoid visualizations that rely on perceptually challenging tasks

comparing angles (4) and 2D areas (5) are harder than position on a common scale (1)

this is why bar charts are preferred over pie charts!

Questions?

(… I'll wait for 2)

For tomorrow:

(07/30 @ 11:30am)

- Lab 18
- Readings
  Matplotlib
- GRQs